

How the ray tracing is combined with the Cloudy model: the simulation-to-spectrum pipeline

pillardisk / transient

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1 Overview

The pipeline turns one Athena++ dump of the perturbed disk into a predicted UV–optical spectrum. It has four stages, implemented in three files under `transient/`, namely `sim_spectrum.py` (stage 1), `sim_cloudy_spectrum.py` (stages 2–4), and `sim_full_spectrum.py` (the thermal and direct components added in stage 4).

1. **Physical fields.** The scale-free simulation cube (density, pressure, velocity on a spherical r, θ, ϕ grid) is anchored to physical units.
2. **Ray tracing.** A central ionizing lamp is placed at the origin. Along every (θ, ϕ) direction the code marches through the simulation’s own density cells and finds where the lamp’s photon budget is exhausted. This defines the three-dimensional ionization front, the irradiated “bowl.”
3. **Cloudy lookup.** Each ray’s irradiated layer is characterized by three numbers (ionizing flux, density, column) and mapped onto a precomputed Cloudy grid, which returns the emitted spectrum of that patch, separately for its illuminated face and its shielded back.
4. **Assembly.** Every patch’s spectrum is weighted by its area, by which face the observer sees, and by a Lambert projection factor, then Doppler shifted by the local gas velocity. The sum, plus the thermal light of the photosphere and the directly seen lamp, is the predicted spectrum.

The key design choice is that Cloudy is **not** run per cell. Cloudy is run once, offline, over a three-parameter family of irradiated slabs. The ray tracing reduces every line of sight from the lamp to a member of that family. This keeps the photoionization physics exact (full radiative transfer within each slab, done by Cloudy) while the geometry, the shadowing, and the kinematics come from the simulation.

2 Stage 1: from simulation to physical fields

The dump (`disk.out1.00012.athdf`) is scale free, so three anchors set the units (all in `CONFIG`):

- $r = 1$ in code units corresponds to $r_0 = 2$ light days, the orbital radius of the disrupted star;

- the black-hole mass is $M_{\text{BH}} = 10^7 M_{\odot}$, which sets the velocity unit $v_0 = \sqrt{GM_{\text{BH}}/r_0} \simeq 5,060 \text{ km s}^{-1}$ and the orbital period at $r = 1$ ($\simeq 740$ days);
- the azimuthally averaged midplane density at $r = 1$ is set to $n_{\text{H}} = 10^{10} \text{ cm}^{-3}$, and the midplane temperature anchor there is 3000 K (the “rescaled” temperature mode, $T \propto P/\rho$, because the simulation’s literal equation of state is numerically inflated).

Everything downstream (columns, recombination rates, Doppler shifts) uses these physical fields.

3 Stage 2: the lamp and the ionization-front march

A lamp-post ionizing source sits at the origin with photon rate

$$Q_{\text{ion}} = 8.5 \times 10^{52} \text{ s}^{-1}, \quad (1)$$

chosen so that the flux arriving at the inner rim is $\log \varphi \simeq 21$, inside the Cloudy grid. Along each of the 256×64 directions (ϕ, θ) the code marches outward through the simulation’s density cells and integrates the recombinations per steradian,

$$S(r) = \int_{r_{\text{in}}}^r n_{\text{H}}^2 \alpha_{\text{B}} r'^2 dr', \quad \alpha_{\text{B}} = 2.59 \times 10^{-13} \text{ cm}^3 \text{ s}^{-1}, \quad (2)$$

comparing it with the available photons per steradian, $Q_{\text{ion}}/4\pi$. The ionization front along that ray is the radius $r_{\text{IF}}(\theta, \phi)$ where $S = Q_{\text{ion}}/4\pi$ (photon counting in Strömgren balance, absorption by the gas the lamp actually shines through). Two outcomes are possible.

- **Radiation bounded.** The budget runs out inside the grid. The ray has a sharp front, and everything beyond it sits in shadow. Dense midplane gas and the arm walls stop the front at small radius, so shadowing is automatic; no separate shadow computation is needed.
- **Matter bounded.** The ray reaches the outer edge of the cube with photons to spare (46% of rays in dump 00012, mostly the dilute layers well above the midplane). The whole column is ionized and transparent.

Note that the photosphere (the $N_{\text{H}} = 10^{24} \text{ cm}^{-2}$ surface) plays no role here. The front emerges from the density field itself, and the two surfaces are different: the ionization front hugs the dense structures much more tightly.

4 Stage 3: reducing each ray to a Cloudy slab

Cloudy models a plane-parallel slab irradiated on one side. Three numbers pick the slab for each ray.

Ionizing flux at the front, with foreshortening. The front is a tilted surface $r = r_{\text{IF}}(\theta, \phi)$. Its tilt relative to the incoming radial rays gives the incidence cosine

$$\mu = \left[1 + \left(\frac{\partial \ln r_{\text{IF}}}{\partial \theta} \right)^2 + \left(\frac{1}{\sin \theta} \frac{\partial \ln r_{\text{IF}}}{\partial \phi} \right)^2 \right]^{-1/2}, \quad \varphi = \frac{Q_{\text{ion}}}{4\pi(r_{\text{IF}}r_0)^2} \mu. \quad (3)$$

A wall that faces the lamp (the inner rim, the arm fronts, where r_{IF} is nearly constant with angle) has $\mu \simeq 1$ and is hit hard. A flared surface climbing steeply with angle is grazed, $\mu \ll 1$. This is the same foreshortening physics as the irradiation temperature in the pillardisk toy model.

Density. The slab’s hydrogen density is the recombination-weighted mean over the ray’s ionized cells, $\log \bar{n}_H = \sum w_i \log n_{H,i} / \sum w_i$ with $w_i = n_H^2 r^2 dr$ evaluated per cell.

Column density. A matter-bounded ray hands Cloudy its true, finite column $N_H = \int n_H dr$, and Cloudy then returns a slab that is transparent to ionizing photons. A radiation-bounded ray is given the grid’s maximum column, and Cloudy’s own internal ionization front terminates the slab, so the near (illuminated) side is fully modeled and the code never has to guess where Cloudy would place the front.

5 The precomputed Cloudy grid

The grid deck (`transient/cloudy/arm_column.in`) uses the same AGN incident spectrum as the older BLR grids (the 15-point `interpolate` table), solar abundances, 100 km s^{-1} turbulence, and varies three parameters:

parameter	range (current grid)	range (<code>arm_column2</code> , queued)
$\log \varphi$ (ionizing photon flux)	17 – 21	17 – 23
$\log n_H$ (hydrogen density)	9 – 12	6 – 12
$\log N_H$ (stopping column)	21.5 – 23.5	21.5 – 24.5

For every grid point the `save continuum` output is split into four channels, each stored as νF_ν per cm^2 of slab surface on a common wavelength grid ($301 - 11,967 \text{ \AA}$):

- `refl_cont`, `refl_line`: continuum and lines leaving the **illuminated face** (the “reflected” emission, toward the lamp);
- `out_cont`, `out_line`: continuum and lines leaving the **shielded back face** (the “outward” emission, away from the lamp).

The four channels are interpolated trilinearly in $(\log \varphi, \log n_H, \log N_H)$ in log flux (`load_grid` in `sim_cloudy_spectrum.py`). Rays whose parameters fall outside the box are clipped to its edge; with the current grid 62% of rays clip at the density floor and 68% at the column edge, which produces the flat uniform patches in the outer disk of the line maps. The queued `arm_column2` run extends the box to remove almost all of this clipping.

6 Face selection and projection

Resonant lines ($\text{Ly}\alpha$, C IV) escape preferentially from the illuminated face, while non-resonant photons leave the shielded back freely. Cloudy’s reflected/outward split encodes exactly this, so the observer-dependent part of the problem reduces to which face of each slab patch the observer sees.

The back-face normal of the front surface is

$$\hat{n} \propto \hat{r} - \frac{\partial \ln r_{\text{IF}}}{\partial \theta} \hat{\theta} - \frac{1}{\sin \theta} \frac{\partial \ln r_{\text{IF}}}{\partial \phi} \hat{\phi}, \quad (4)$$

and with the observer direction $\hat{o}$ (inclination i from the disk axis, plus an azimuth in the phase sweeps) the blend

$$\cos v = \hat{n} \cdot \hat{o}, \quad s_{\text{out}} = \frac{1}{2} [1 + \tanh(\cos v / 0.3)] \quad (5)$$

weights the two faces: $s_{\text{out}} \rightarrow 1$ shows the shielded back (patch between lamp and observer), $s_{\text{out}} \rightarrow 0$ shows the lit face, and the smooth blend avoids a hard switch for edge-on patches. An optically thick patch is also a Lambert emitter, so its flux carries the projection factor $4|\cos v|$ (the factor 4 makes an isotropic-equivalent luminosity, so a face-on patch is brighter and an edge-on patch fainter than the isotropic average, energy conserved over viewing angles). Matter-bounded (transparent) patches emit both channels isotropically with no projection factor. At $i = 45^\circ$ half the radiation-bounded patches show the observer their lit face and half their shadow face.

7 Stage 4: assembling the spectrum

Each ray's patch has surface area

$$dA = \frac{(r_{\text{IF}} r_0)^2 d\Omega}{\mu}, \quad (6)$$

the solid-angle patch enlarged by $1/\mu$ for the tilt, which together with $\varphi \propto \mu$ conserves the photon budget per steradian. Because the observer sees only one face of an opaque disk, only the observer-side (upper, $\theta \leq 90^\circ$) hemisphere enters the reprocessed sum, matching the one-sided thermal treatment below; the lower-hemisphere patches have dA set to zero at the source, so the spectra, both sweeps, and the line maps all inherit the restriction consistently. The luminosity spectrum is then

$$L_\lambda = \sum_{\text{rays}} dA \left[w_{\text{ref}} F_\lambda^{\text{refl}} + w_{\text{out}} F_\lambda^{\text{out}} \right]_{\lambda \rightarrow \lambda(1-v_{\text{los}}/c)} + 4 \cos i L_{\lambda, \text{th}}^{\text{base}} + L_{\lambda, \text{AGN}}, \quad (7)$$

with the three pieces as follows.

- **Reprocessed light** (the Cloudy sum). Each patch's entire spectrum, continuum and lines alike, is Doppler shifted by the recombination-weighted line-of-sight velocity of its ionized cells, taken from the simulation velocity field. This is what broadens the lines into the observed $\pm 3000 \text{ km s}^{-1}$ double-horned profiles and smears the Balmer jump; it is not applied artificially to lines only.
- **Thermal light.** Every disk column radiates as a blackbody at the temperature sampled at its own photosphere, the depth where the vertical column from the wedge surface reaches $N_{\text{H}} = 10^{24} \text{ cm}^{-2}$. This single component contains the viscous disk, the TDE-heated region, and the arms, because they are all part of the one simulation temperature field; the $4 \cos i$ factor is the same Lambert convention as above.
- **Direct AGN light.** The lamp seen directly, using the identical `interpolate` SED that irradiates the Cloudy slabs, normalized so that its photon rate above 1 Ryd equals Q_{ion} . Input SED, reprocessed emission, and direct light therefore share one spectral shape and one normalization.

The inclination sweep recomputes only face selection, projection, and Doppler geometry per inclination (the march and the Cloudy lookup are inclination independent); the phase sweep does the same per observer azimuth at fixed inclination.

8 The line maps and the self-consistency check

The face-on line maps (CIV, Mg II, H α , and the RGB composite) are **not** an input to the spectra. Both are projections of the same per-ray line luminosities. The spectrum sums them in wavelength; the map paints each ray’s line luminosity back along its ionized cells with the recombination weight $n_{\text{H}}^2 r^2 dr$ (with the front surface interpolated in θ so the rendering shows the continuous bowl rather than the 64 discrete polar rays).

Numerical check, dump 00012 at $i = 45^\circ$ (upper hemisphere). Summing the H α map gives 1.83×10^{40} erg s $^{-1}$, and integrating the line channel of the 45° spectrum over the full Doppler-broadened profile (6350 – 6800 Å) returns the same number, because they are the same photons, projected spatially in one case and spectrally in the other. (A narrow window such as 6510 – 6620 Å underestimates the spectral side by a factor of a few, since the ± 3000 km s $^{-1}$ horns carry the flux outside it.)

9 Approximations worth remembering

1. The march uses a single case-B recombination coefficient at 10^4 K; Cloudy then handles the real thermal and ionization structure inside each slab, but the location of the front itself is photon counting only, with no diffuse-field transport between rays.
2. Slabs are characterized by one mean density and one column per ray; density gradients along the ionized segment are folded into the recombination-weighted mean.
3. First-order Doppler shift only; relativistic beaming ($\sim 20\%$ on the inner rim) and gravitational redshift ($\sim 1\%$) are neglected.
4. No patch-on-patch occultation along the observer’s line of sight, so the sweeps stop at $i = 80^\circ$. For the same reason the far-side strip of the lower rim wall that is really visible through the central funnel is neglected (the reprocessed sum keeps the upper hemisphere only).
5. The lamp is a point at the origin. For $M_{\text{BH}} = 10^7 M_\odot$ the gravitational radius is 5.7×10^{-4} light days, so any standard lamp height of $6 - 20 r_g$ changes the incidence angles at the simulated radii (≥ 1 light day) by well under a degree; the physically load-bearing part of an elevated lamp, its clear view over the inner disk, is present as the assumption that the unsimulated hole ($r < 1$ light day) is transparent.
6. 62% / 68% of rays currently clip at the Cloudy grid’s density floor / column edge; superseded once `arm_column2` is run and extracted (the pipeline adopts the extended cache automatically).